

Critical Research Review: Ambiguity-Aware LLM Post-Training

Subject: Comparative Analysis of “Teaching LLMs to Clarify or Act” (Sutthiboriban et al.) vs. Our Project (Ask-Before-Answer)

1. Experimental Setup

Both projects tackle the “clarify-or-act” decision boundary using the Qwen2.5-7B base architecture and the AmbigNQ/AmbigQA dataset. Both employ Supervised Fine-Tuning (SFT) to establish structural behavior (outputting actions, facets, and responses) followed by preference optimization.

Key Methodological Differences:

- * **Data Curation:** The competitor focused heavily on data quality scaling. They generated datasets using a 7B model (for base SFT) and a 32B model (for “Alt-SFT” which included explicit reasoning traces). For DPO, they contrasted synthetic negatives against “hard negatives” mined by finding instances where the 7B model disagreed with the 32B model.
- * **Post-Training Algorithms:** The competitor restricted their analysis to SFT and DPO. Our methodology is significantly broader, evaluating SFT, DPO, ORPO, and introducing **GRPO (Group Relative Policy Optimization)**.
- * **Evaluation Framework:** The competitor ultimately dropped their LLM-as-a-Judge and relied on deterministic string/format matching for their metrics. We employed a rigorous, multi-dimensional LLM-as-a-Judge framework (google/gemma-2-2b-it) evaluating 9 distinct metrics, separating structural compliance from semantic usefulness.

2. Quantitative Performance

Note: Raw metric comparison is only suggestive due to the competitor dropping the LLM judge, which likely inflated their raw scores compared to our strict semantic evaluation.

- **Base Model:** The competitor’s Qwen Base achieved 0.182 Answer Accuracy, whereas ours achieved 0.050. This discrepancy is almost certainly due to evaluation criteria—our LLM judge strictly evaluates semantic correctness, whereas regex matching often yields false positives.
- **SFT Impact:** Both projects found SFT to be wildly successful at establishing basic behavior. The competitor saw Macro F1 jump from 0.52 to 0.63. We saw Macro F1 jump from 0.464 to 0.556. Both projects observed that SFT drastically corrects the base model’s over-clarification bias (Competitor Clarify Ratio: 1.41 $\rightarrow$ 0.92; Our Clarify Ratio: 1.56 $\rightarrow$ 1.30).
- **Preference Optimization:** The competitor’s DPO pipelines yielded mixed results, generally dropping Answer Accuracy (e.g., Alt-SFT-DPO dropped from 0.242 to 0.173) while causing the model to over-clarify (Clarify Ratio jumped back to 1.29). Our DPO pipeline exhibited a similar

but more extreme pathology: it maximized structural metrics (Macro F1: 0.582, Clarify Ratio: 1.16) but absolutely destroyed Answer Accuracy (0.000).

- **The GRPO Advantage:** Because the competitor stopped at DPO, they have no equivalent to our GRPO results. Our GRPO pipeline (u8xjwcjv) achieved an Answer Accuracy of 0.150 (tripling our SFT baseline) while maintaining a perfect Facet Generation Rate (1.0), demonstrating a solution to the DPO degradation problem.

3. Ablation and Training Effects

Both projects independently reached a highly similar conclusion regarding DPO: **DPO is a behavior amplifier that is hyper-sensitive to the dataset, often eroding semantic quality.**

- **Competitor Evidence:** They argue that DPO “does not uniformly improve performance; instead it sharpens whichever tendencies are encoded in its preference pairs.” When they introduced hard negatives, their model became “strongly clarification-heavy,” avoiding answers unless extremely confident.
- **Our Evidence:** Our results provide stronger evidence as to *why* this happens. Our DPO model achieved the best Action F1 (0.457) but an Answer Accuracy of 0.000. This proves the model learns to “game” the DPO objective by perfecting the structural classification task (Clarify vs. Answer) while losing the generative semantic grounding required to actually answer the question.

4. Evaluation Validity

Our evaluation framework is demonstrably superior and more valid. The competitor explicitly states in their Author Contributions: “*LLM Judge was later dropped as the project direction evolved.*” Consequently, their “Clarification Quality” and “Answer Accuracy” metrics are likely reliant on ROUGE/BLEU or exact-match heuristics.

In contrast, clarify-or-act is fundamentally a semantic task. Generating a question like “*What do you mean?*” is structurally a clarification, but semantically useless. Our 9-metric framework, powered by **gemma-2-2b-it**, evaluates *Usefulness*, *Ambiguity Detection F1*, and fuzzy *Answer Accuracy*. This provides a much more valid representation of real-world conversational agent performance.

5. Fairness of Comparison

Comparisons of raw Answer Accuracy or F1 scores between the two projects are **not directly comparable** due to the divergence in evaluation methodology (Heuristic vs. LLM-as-a-Judge).

We confirmed this disparity by directly evaluating the competitor’s model (downloaded from Huggingface and labeled as `clarifier_lora` in our ablation study) against our rigorous, semantic LLM-as-a-judge framework. Under our framework, the competitor model reveals significant flaws: * **Clarify Recall of 1.0, but a Clarify Ratio of 1.66:** The model drastically over-clarifies, asking questions for nearly every input regardless of ambiguity. * **Answer Accuracy of 0.000:** The model completely fails to produce semantically correct, context-grounded answers. * **Facet Generation Rate of 0.000:** The model fails to produce the required JSON-formatted facets needed for structured reasoning.

This direct head-to-head comparison proves that while the competitor’s heuristic metrics suggested strong performance, their model collapses when evaluated strictly for semantic correctness and structural compliance. Our framework is significantly more demanding, proving that our absolute scores (e.g., GRPO’s 0.150 Answer Accuracy) reflect genuine semantic capability rather than heuristic formatting.

However, the **relative trends** remain highly comparable and cross-validate each other. Both projects definitively prove that SFT is required for structural alignment, and both projects expose the brittleness of standard DPO when applied to multi-objective reasoning tasks.

6. Statistical and Experimental Rigor

- **Weaknesses in Both:** Neither project reports variance, standard deviations, or confidence intervals computed across multiple random seeds. In RLHF/DPO research, policy optimization is notoriously unstable, making seed variance critical for establishing statistical significance.
- **Competitor Strength:** They conducted a highly rigorous data ablation, isolating the effects of 7B-synthetic vs. 32B-synthetic vs. hard-mined negatives.
- **Our Strength:** We conducted a mathematically rigorous hyper-parameter sweep (Bayesian optimization with Hyperband early stopping) for our RL stage. The competitor does not report extensive hyper-parameter tuning for their DPO margins or learning rates.

7. Strengths and Weaknesses

Competitor Strengths: * Sophisticated data engineering pipeline (using a 32B teacher to mine hard negatives from a 7B student). * Strong analysis of how specific negative pair formulations bias the resulting DPO policy.

Our Strengths: * Far more comprehensive algorithmic suite (SFT, DPO, ORPO, GRPO). * Superior evaluation framework (LLM-as-a-Judge assessing semantic usefulness). * We present a *solution* (GRPO) to the DPO degradation problem, whereas the competitor simply observes the degradation.

8. Scientific Interpretation

The combined results of both projects suggest a unified theory of post-training for ambiguity resolution: 1. **SFT is necessary and highly effective** for teaching the structural format of facet-based reasoning. 2. **Offline Preference Optimization (DPO/ORPO) is fundamentally misaligned** with the clarify-or-act task. Because ambiguity resolution requires a delicate balance of caution (clarifying) and helpfulness (answering), static preference pairs force the model to over-index on whichever behavior the dataset implicitly favors, leading to reward hacking (e.g., our DPO model’s 0.00 Answer Accuracy). 3. **Online Reinforcement Learning (GRPO) is the superior paradigm.** By replacing static preference pairs with dynamic, programmatic reward functions (Accuracy, Formatting, KL-penalty), GRPO allows the model to explore and find the optimal Pareto frontier between structural compliance and generative accuracy.

9. Novelty and Contribution

Our project provides a substantial contribution beyond the competitor’s work. While the competitor successfully identifies the limitations of DPO for this task, they attempt to solve it purely through dataset engineering (which only yielded marginal/mixed results).

Our work introduces **GRPO** to the clarify-or-act problem. To our knowledge, demonstrating that online RL with programmatic rewards can successfully train ambiguity-aware agents—bypassing the need for expensive, human-annotated, or 32B-mined preference pairs—is a highly novel and impactful empirical claim.

10. Publication-Quality Assessment

From a peer-review perspective, our manuscript is highly competitive.

Strongest Evidence: Our ablation study clearly isolates the degradation caused by DPO and the subsequent recovery/triumph of GRPO. The use of Bayesian hyper-parameter sweeps adds significant engineering credibility. **Primary Threat to Validity:** The lack of multi-seed variance reporting. Without confidence intervals, reviewers may question if GRPO’s superiority is statistically significant or the result of a “lucky seed.” **Recommended Additional Analysis:** To unequivocally dominate the competitor’s approach, we should run one final DPO baseline using *their* proposed “hard negative mining” strategy. If our GRPO pipeline still outperforms a heavily optimized, hard-negative DPO model, the claim that online RL > offline PO for ambiguity resolution becomes practically unassailable.